

Suyeon Lee

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RESEARCH INTEREST

My research focuses on multimodal AI, with a core emphasis on audio-visual speech and source separation. I investigate how cross-modal cues can be leveraged to isolate target signals and enhance auditory clarity in complex acoustic environments. My work aims to improve communication experiences through robust, human-centric scene understanding.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. in Electrical Engineering</i>	Mar 2025 - Feb 2028 (Expected) <i>Advisor: Prof. Joon Son Chung</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>M.S. in Electrical Engineering</i> Thesis: Multi-modal Speech and Audio Separation System	Mar 2023 - Feb 2025 <i>Advisor: Prof. Joon Son Chung</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>B.S. in Electrical Engineering</i>	Mar 2018 - Feb 2023

EXPERIENCE

Undergraduate Research Intern <i>Multimodal AI Lab, KAIST</i> - Acquired foundational knowledge in generative models, specifically exploring diffusion-based models - Implemented active speaker detection baseline	Mar 2022 - Feb 2023 <i>Advisor: Prof. Joon Son Chung</i>
Research Intern <i>AIRS Company, Hyundai Motor Group</i> Developed wake word detection baseline model for in-car voice assistant	Sep 2021 - Feb 2022 <i>Seoul, South Korea</i>
Undergraduate Research Intern <i>SSSC Lab, KAIST</i> Applied fundamental machine learning principles to develop a baseline speech recognition model utilizing PyTorch.	Mar 2021 - Aug 2021 <i>Advisor: Prof. Hoirin Kim</i>

HONORS AND AWARDS

NIST Speaker Recognition Evaluation (SRE) <i>1st Place in the Audio Track and 4th Place in the Audio-Visual Track</i>	2024 <i>San Juan, Puerto Rico</i>
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SKILLS

Programming: PyTorch, Python, C
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ACADEMIC ACTIVITIES

Teaching Assistant <i>KAIST - EE488: Deep learning for Computer Vision</i> Supported students in coding assignments by offering one-on-one debugging and technical guidance	Fall 2023, Fall 2025
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Korean: Native

English: Proficient

- [6] **Plug-and-Steer: Decoupling Separation and Selection in Audio-Visual Target Speaker Extraction**
Doyeop Kwak*, **Suyeon Lee***, Joon Son Chung
Preprint
- [5] **Cinematic Audio Source Separation Using Visual Cues**
Kang Zhang*, **Suyeon Lee***, Arda Senocak, Joon Son Chung
IEEE / CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- [4] **Model-Guided Dual-Role Alignment for High-Fidelity Open-Domain Video-to-Audio Generation**
Kang Zhang*, Trung X. Pham*, **Suyeon Lee**, Axi Niu, Arda Senocak, Joon Son Chung
Conference on Neural Information Processing Systems (NeurIPS) 2025
- [3] **FlowAVSE: Efficient audio-visual speech enhancement with conditional flow matching**
Chaeyoung Jung, **Suyeon Lee**, Ji-Hoon Kim, Joon Son Chung
Interspeech 2024
- [2] **Seeing through the conversation: Audio-visual speech separation based on diffusion model**
Suyeon Lee*, Chaeyoung Jung*, Youngjoon Jang, Jaehun Kim, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024
- [1] **TalkNCE: Improving active speaker detection with talk-aware contrastive learning**
Chaeyoung Jung*, **Suyeon Lee***, Kihyun Nam, Kyeongha Rho, You Jin Kim, Youngjoon Jang, Joon Son Chung
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024